
BIOLOGY

9700/53

Paper 5 Planning, Analysis and Evaluation

October/November 2017

MARK SCHEME

Maximum Mark: 30

Published

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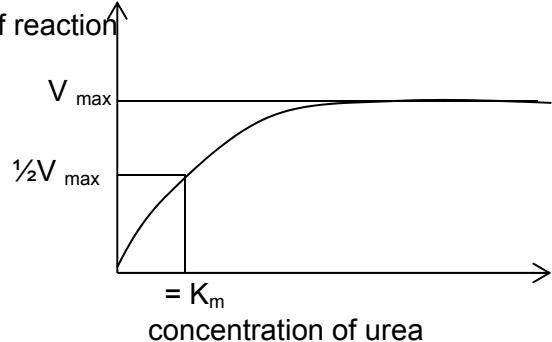


Mark scheme abbreviations

;	separates marking points
/	alternative answers for the same point
R	reject
A	accept (for answers correctly cued by the question, or by extra guidance)
AW	alternative wording (where responses vary more than usual)
underline	actual word given must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ora	or reverse argument
mp	marking point (with relevant number)
ecf	error carried forward
I	ignore
AVP	alternative valid point

Question	Answer	Marks
1(a)(i)	weigh / measure (out) / take / put / add / use/ AW, 6 (.01) g urea ; add 500 cm ³ (distilled / deionised) water (and stir until dissolved) ;	2
1(a)(ii)	<i>idea of</i> removing (a known and) same volume of urea solution (with second dilution taken from first dilution, etc.) at each stage of dilution ; <i>idea of</i> adding (a known and) same volume of (distilled) water at each stage of dilution ; A as a diagram showing sequence of dilution A as a table showing volumes (and concentrations) A proportional dilution for max 1 (ecf) if 4 dilutions correctly gained	2
1(b)(i)	<i>independent</i> temperature ; <i>dependent</i> conductivity (of enzyme and substrate / ions / solution) ; A in conductivity units	2
1(b)(ii)	substituting, the active enzyme / urease, by an unreactive substance (at all temperatures) A e.g. boiled or denatured enzyme / water in place of enzyme / 0.0M enzyme / (solution of) urea without urease or substituting, urea / substrate, with water ; I 'use (distilled) water' unqualified A e.g. water in place of substrate / 0.0M substrate / (solution of) urease without urea	1

Question	Answer	Marks
1(b)(iii)	<i>max 5 if mp7 not given</i> <i>any 6 from:</i> 1 <i>ref. to a suitable range of at least 5 temperatures ; A any 5 in the range 10–70 °C</i> 2 <i>ref. to using suitable apparatus (to incubate enzyme and urea solutions at constant temperature(s)) ; e.g. (thermostatically controlled) water baths / incubators / thermostatically or temperature controlled room / (magnetic stirrer) hotplate</i> A beaker plus hot water as water-bath I air conditioning 3 <i>ref. to using same volume of urease each time ; total volume must not exceed 30 cm³</i> 4 <i>ref. to using same volume of (each) urea concentration ; total volume must not exceed 30 cm³</i> 5 <i>ref. to using (same volume of) buffer to maintain a constant pH ;</i> A use buffer to control pH <i>procedure:</i> 6 <i>ref. to incubating urease and urea concentrations separately ;</i> 7 <i>ref. to mixing urea and urease solutions (on the magnetic stirrer) and immersing (conductivity) probe ;</i> I 'probe is used' unqualified 8 <i>ref. to taking reading (from meter) at same time (for each solution / temperature) ;</i> A any stated time from 0 s to 5 min A take reading immediately / AW A 'use meter to measure rate of reaction' if time context correct 9 <i>ref. to testing each of the concentrations (of urea) at each temperature ;</i> 10 <i>ref. to a min. of 3 replicates / repeats and a mean / find anomalies ;</i>	6

Question	Answer	Marks
1(b)(iii)	11 <i>ref. to suitable hazard and precaution ;</i> A low risk experiment I medium risk experiment R no risk / high risk A urease / enzyme is allergen or irritant and wear gloves / goggles A Ammonia (given off at high temperature) is irritant / mask or gloves	
1(c)(i)	1 one curve drawn correctly ; <i>does not need to go through origin</i> A straight line up then level without curving 2 $V_{\max}$ shown ; 3 $\frac{1}{2}V_{\max}$ shown ;  = K_m concentration of urea 4 K_m shown ; A Michaelis-Menten (constant)	4
1(c)(ii)	1 D ; 2 (temperature showing) lowest K_m or shows, greater / greatest / higher / highest, affinity (of the, enzyme / urease) for its substrate / AW ;	2

Question	Answer	Marks
2(a)(i)	<i>any 1 from:</i> 1 time of soaking (grain in the salt solutions) / 12 hours soaking for each set / AW ; 2 number of grains (soaked in each, salt concentration / set / covered petri dish) ; 3 temperature (of germination / incubation) ; <i>20 °C if quoted</i> 4 time intervals of recording (germination) ; A recorded at 8 hour intervals or recorded over 5 days 5 <i>idea of</i> taking a standard appearance of grain when judged to be germinated ; e.g. <i>emergence of radicle</i>	1
2(a)(ii)	<i>any 1 from:</i> 1 volume of (salt) solution used, on the filter paper / in the (Petri) dish ; I 'amount' unqualified 2 age of grain ; 3 (use) undamaged / not infected / not diseased / AW, grain ; I size / mass, of grain 4 <i>idea of</i> light (exposure of grains during germination / breaking dormancy) ; 5 supply of, air / oxygen (to the grain) ; 6 <i>idea of</i> spacing of grains ; 7 pH (of solution) ;	1

Question	Answer						Marks	
2(b)(i)	concentration of salt solution /mmoldm ⁻³		mean (cumulative) percentage of grains that had germinated each day				2	
			1		5			
			X	Y	X	Y		
			0	93.8	92.0	98.0		97.0
			10	93.0	90.2	97.4		96.4
			20	92.2	87.0 ;	96.6		93.6
			30	96.4 ;	90.0	96.4		93.0
			40	92.0	90.4	95.4		92.4
			50	91.3	91.0	95.0		91.6
			60	91.2	90.0	95.0		91.0
2(b)(ii)	1 Pearson's linear correlation ; A Pearson's R Pearson's Rank correlation						2	
	2 data collected is continuous							
	or data, is / seems to be, (from a population that is) normally distributed							
	or data / results, appears to be linear ;							

Question	Answer	Marks
2(c)	<i>I data quotes unqualified</i> any 3 from: 1 <i>idea that</i> (germination is rapid because) nearly all / approx. 90% / majority / most, of grains, germinated during the first day ; 2 <i>idea that</i> (control shows) some of the barley, will not germinate, over the period of the experiment / in the first five days ; 3 percentage germination (generally), decreases as concentration of salt increases / increases as concentration of salt decreases ; A <i>idea of negative</i> correlation 4 more grains have germinated after 5 days (than after 1 day) / ora ;	3
2(d)(i)	<i>must be comparative</i> <i>I data quotes unqualified</i> germination of barley is, higher / highest, in X (than Y at, all salt concentrations / every value / stated value(s) from 30 mmol dm⁻³ to 60 mmol dm⁻³) ;	1
2(d)(ii)	any 1 from: 1 <i>idea of</i> measuring / recording / investigating / AW, germination of, X and Y / both, in salty <u>soil</u> ; 2 <i>idea of</i> measuring / recording / investigating / AW, growth of, X and Y / both, in salty <u>soil</u> ; 3 <i>idea of</i> measuring / recording / investigating / AW, yield of, X and Y / both, in salty <u>soil</u> ; 4 AVP ; e.g. a field investigation involving a transect across an area from low to high salt then measuring abundance of the types along the transect	1